

Linear algebra: Exercise session 1

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1. For the following vectors:

$$v = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}.$$

- (a) Compute $3v$ (scalar multiplication) and $3v - w$ (linear combination);
- (b) Compute the length of w and v and check the triangle and the Cauchy Schwartz inequalities on v and w ;
- (c) Compute $\|v\|^2 - \|w\|^2 + \|4v - 2w\| - \langle v + w, v - w \rangle$.

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2. Are vectors $a = (1, -1)$ and $b = (2, 0)$ linearly independent? Why? Are they a basis for $\mathbb{R}^2$?

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3. Are vectors $a = (1, 3, 0)$ and $b = (1, 0, 2)$ linearly independent? Why? Are they generating $\mathbb{R}^3$?

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4. Are vectors $a = (1, 3, 0)$, $b = (1, 0, 2)$, and $c = (3, 3, 4)$ linearly independent? Why? Are they a basis for $\mathbb{R}^3$?

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5. If vector a is perpendicular to vectors b and c . Is it also perpendicular to $b + 2c$? Are b and c linearly dependent?

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6. If a and b are perpendicular unit vectors. Show that $\|a - b\|^2 = 2$.

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7. Compute the angle value between $a = (1, 0, -\sqrt{3})$ and $b = (3, 0, 0)$.

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8. Write a MATLAB code that prompts the user to enter a scalar as the dimension of a vector space and displays the elementary, zeros and ones vectors in this space.

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9. Write a MATLAB code that prompts the user to enter a vector and a scalar, then displays the vector, the dimension of vector space containing the vector, the length (norm) of the vector, and the scalar multiplication of the scalar with the vector.

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10. Write a MATLAB code that prompts the user to enter two vectors, then displays the addition and the inner product of them, and the angle value between them.
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